Subsalt Takehome Assignment

Executive Summary

Our modeling team built a model to identify which customers are most likely to churn (defined as making no purchase in the 90 days following a reference cutoff date) so the marketing team can focus its limited budget where it's most valuable. Using the developed model, if we rank every customer by churn risk and only contact the highest-risk 10%, we expect roughly 87% of those customers to churn. This represents a 1.8x improvement over contacting customers at random meaning the marketing team can spend the same budget and reach nearly twice as many real churners.

Data Processing & Churn Definition

We defined churn using a 90-day window as a starting point for this analysis. Any customer making a purchase after this cutoff (unless it was their first purchase) is considered a non-churn customer. This was checked against the median inter-purchase gap (~56 days for repeat customers with a standard deviation of ~84 days) to show the cutoff is somewhat arbitrary. No cutoff cleanly separates lapsed customers, but 90 days is past the typical reorder cycle. It was also built as an adjustable parameter so the model can easily be re-tuned and evaluated.

To create the modeling dataset we cleaned and processed the data through a few steps:

Filters

- Removed non-standard transactions and returns as they do not represent typical customer interactions
- Removed any customers who only purchased once (prior to the cutoff) as they do not have enough purchase history to be considered churned
- Removed any customers who made their first purchase after the 90-day cutoff

Features developed for modeling

- **Purchase frequency** features such as recency, tenure, number of purchases, and deceleration (last purchase gap duration over median purchase gap duration) were built to represent a customer's purchasing habits
- **Purchase size** features such as total revenue, average order value, and average basket size were built to represent purchasing choices

Overall our final dataset contained 3,682 customers across 28,457 purchases representing $13.5M. Of those customers, ~47% were considered as having churned.

Modeling Overview

To model customer churn we built two models: a Logistic Regression to establish a baseline performance and a Random Forest. These were evaluated using AUC and precision@top-k. The evaluation metrics were chosen due to their utility for the marketing team because they represent rank-order performance. Understanding the precision of our model on the top 10% highest-risk predictions, we can understand how much lift we can expect compared to a random sample and identify high-risk individuals for outreach.

Pre-processing & Modeling Details

- We used a train-test split of 80%-20% respectively using cross-validation for training and hyperparameter tuning
- We dropped highly correlated features (correlation >=0.85). In this analysis this only included total units (correlated with total revenue)
- We log-transformed skewed variables for the logistic regression. This included total revenue, average order value, average basket size, number of purchases, and the last gap over median gap
- We also standardized features for the logistic regression
- The random forest hyperparameters (number of estimators, max depth, and minimum samples per-leaf) were optimized using a grid search scoring against AUC

Model Performance

Below is a table showing model performance across the two models as measured against the hold-out dataset:

Metric	Logistic Regression	Random Forest
Area Under the Curve (AUC)	0.785	**0.795**
Precision@top-10%	0.797 (1.71x lift)	**0.865 (1.86x lift)**
Precision@top-20%	**0.791 (1.70x lift)**	0.784 (1.68x lift)
Precision@top-30%	0.752 (1.62x lift)	**0.766 (1.65x lift)**
F1 - Weighted Average	0.712	**0.719**

Overall, the random forest performed the best, specifically at the top decile. This is the metric that represents the most value to the marketing team for prioritizing outreach so our recommendation is to use the random forest model for identifying high-risk customers.

Future Improvements & Decision Confidence

I’m most confident in the modeling and evaluation decisions, especially given the timeline. While more models could have been included with more sophisticated hyperparameter tuning, the metric decisions of AUC and precision@top-k are able to directly inform downstream marketing decisions.

I’m confident in the high-level processing decisions, however in a real-world scenario additional context would be helpful in deciding cutoffs, customer segments, and other high-value adjustments that could be incorporated into the processing and feature engineering steps in the analysis.

The definition of churn is complex and was the most difficult decision. The 90-day cutoff is a simple definition that was easy given the time constraints, but in a real-world scenario would likely take the bulk of the preparation time. I am also least confident in the binary definition of churn as it does not consider customer segments. It would be best to have separate models representing wholesale customers, retail customers, and seasonal customers. This would also help the marketing team customize their outreach to specific segments.

Ideas for Future Improvements

I identified a few improvement ideas while working through this exercise:

- **Incorporating external data:** Country economic data could help to inform spending habits by customers and act as a leading indicator for future churn
- **Customer segmentation:** Creating customer groups based on purchasing behavior and using those groups as features in the model could help separate high-value vs low-value customers for marketing outreach
- **Sequence-based Analysis:** The current approach uses customer-level data with derived time features. A sequence based analysis (survival analysis, LSTM, etc.) could help incorporate more of the temporal signal and better identify customers likely to churn